PERFORMING RIGS

AutoPilot Firmware Validation Report

Document	AP-VVR-003
Revision	Rev A
Product	AutoPilot
Hardware Revision	Hardware Revision 1 (bench / field generation) — only HW available for this session
Firmware Revision	Autopilot_firm05-08 · 915 MHz · UI AutoPilot V01 R39 · Remote V01 R42
Validation Date	August 6, 2026
Prepared By	Nathan Destro, Performing Rigs
Location	Bench / shop

Validation Summary

Validation Status	OPEN
Functional Result	CONDITIONAL PASS

Validation Status — Validation remains open. Intentional trigger recognition on this firmware is strong, and the AP-VVR-002 idle / post-reset false-trigger pattern was not observed during dedicated idle watches. However, Relay Contact mode produced continuous unintended triggering under specific connected-but-inactive / floating bench-supply conditions. In addition, this session used Hardware Revision 1 only; it does not by itself clear Hardware Revision 2 production units.

Functional Result — Relay Contact and 12 V / 24 V intentional triggers met count targets with zero misses under intentional-fire tests. Diagnostic monitoring no longer jumps immediately to RUNNING. These functional results do not constitute Batch-2 engineering release approval.

1. Executive Summary

Firmware package Autopilot_firm05-08 (Americas 915 MHz) was flashed and validated against the AP-VVR-003 test matrix derived from AP-VVR-002 remaining work and Noxon's latest trigger-mitigation claims.

Positive results. Cold-start, soft-reset, and power-cycle idle watches (T-01–T-03) and extended soak (T-13) passed with no spontaneous false triggers. Intentional Relay Contact (18/18), 12 V (12/12), and 24 V (12/12) fires all succeeded with zero misses. INFO / monitoring remained usable during signaling (standby and live mV readouts captured). Cross-mode discipline improved versus AP-VVR-002 F-004: a dry-contact pulse while in 12/24 V mode produced no action; 9 V in 12/24 V mode also produced no activation.

Critical concern. In Relay Contact mode, with a bench supply attached to the trigger input but not commanding an intentional voltage (and again with a 12 V supply connected but not activated), AutoPilot continuously false-triggered. Observed trigger-input readout during the fault was approximately 1909–1958 mV, with occasional brief ~16 mV readings. Applying 9 V raised the readout to approximately 2696–2719 mV. The same 0 V / 9 V conditions in 12/24 V mode did not false-trigger and did not activate on 9 V.

Hardware scope limit. Only Revision 1 hardware was available. Batch-2 units use revised trigger circuitry. Firmware behavior on Rev 1 is necessary evidence for field Rev 1 support, but is insufficient alone to close engineering release for Rev 2 shipment.

2. Evidence Summary

Validation Item	Result
T-01 Cold power-on idle (false-trigger watch)	PASS
T-02 Soft reset / reboot idle watch	PASS
T-03 Full power-cycle idle watch	PASS
T-04 Idle with nearby motion / noise	NOT TESTED (no setup)
T-05 Relay Contact intentional fires	PASS (18/18, 0 misses)
T-06 12 V intentional fires	PASS (12/12, 0 misses)
T-07 24 V intentional fires	PASS (12/12, 0 misses)
T-08 9 V / mode cross-check	FAIL (Relay Contact false trigger) — see F-001
T-09 Monitoring / INFO during signal	PASS (no forced jump to RUNNING)
T-10 Startup / minimum current	NOT TESTED (no measurement means)
T-11 Threshold / noise discrimination (operator judgment)	PASS
T-12 Rev 1 vs Rev 2 same firmware	NOT TESTED (Rev 1 only)
T-13 Long idle soak	PASS
T-14 Mixed intentional + idle	PASS (combined with session)
Relay pulse while in 12/24 V mode	NO ACTION (expected / improved isolation)
False trigger — Relay mode + inactive connected supply	OBSERVED (continuous)

Trigger-input voltage observations (INFO / monitor)

Condition	Approximate readout
Standby	1664–1670 mV
Dry contact (active)	1922–1990 mV
Fault window (Relay mode, supply connected / not activated)	1909–1958 mV (continuous false triggers); occasional ~16 mV
9 V applied	2696–2719 mV
12 V applied	2806–2819 mV
24 V applied	3045–3051 mV

Noxon thresholds (Iker, 2026-08-06): Relay trip ≈ 1850 mV (clean contact ≈ 1950 mV); 12/24 INFO ≈ 2400 mV and 800 mV; applied ~6–8 V polarity-dependent for voltage modes; line pre-energized ~3 V for relay sense. Bench fault band 1909–1958 mV sits above the relay trip — aligned with F-001. Full comparison: Iker-Trigger-Thresholds-2026-08.md.

3. Findings

F-001 — Continuous false trigger in Relay Contact mode with inactive / floating bench supply

While configured for Relay Contact mode, with a bench power supply connected to the trigger input but not sending an intentional commanded voltage (and similarly with a 12 V supply connected but not activated), AutoPilot continuously initiated triggers. Resetting the unit did not clear the condition: it re-activated immediately.

During the fault, the trigger-input readout remained near 1.91–1.96 V, overlapping the same band observed for intentional dry-contact (~1.92–1.99 V). That suggests the input was sitting in the "contact closed / active" voltage region without an intentional closure.

Aligned with Iker's thresholds (2026-08-06): relay trip ≈ 1850 mV, clean contact ≈ 1950 mV. The fault window sits above the trip — continuous fire is expected for that INFO band. Raising the relay trip slightly is Noxon's suggested mitigation, with long-cable margin caution.

The same electrical conditions in 12/24 V mode did not produce false triggers.

Engineering Impact

- Unexpected motion from Relay Contact mode under a connected-but-idle voltage source is a critical production hazard if that wiring state can occur on set.
- May indicate insufficient discrimination between open / floating / leakage and true dry contact, or unsafe behavior when a voltage source is paralleled onto the relay input.
- Requires Noxon root-cause analysis and corrective action before Relay Contact can be considered validated on this firmware.

EMS: AP-001 (related / expanded)

Severity: Critical

Status: OPEN

F-002 — Idle / post-reset spontaneous false triggers not observed (this session)

Dedicated idle watches after cold start, soft reset, and full power cycle, plus extended soak, showed no spontaneous false triggers of the AP-VVR-002 F-002 pattern (~30 s post-reset).

Severity: Info

Status: Not reproduced this session — does not close F-001; does not by itself close historical AP-001 for all environments

F-003 — Intentional trigger recognition met targets

Mode	Count	Misses
Relay C ontact	18	0
12 V	12	0
24 V	12	0

Severity: Info

Status: Validation Passed (intentional-fire scope)

F-004 — Mode isolation improved vs AP-VVR-002 F-004

- Relay / dry-contact pulse while in 12/24 V mode: no action.
- 9 V while in 12/24 V mode: no activation (contrast with AP-VVR-002, where 9 V fired in both modes).
- 9 V in Relay Contact mode: input rises to ~2.7 V (see voltage table); interaction with F-001 wiring must be interpreted carefully.

Severity: Info / Low

Status: Behavior change documented — confirm with Noxon whether intentional

F-005 — Monitoring / INFO usable during signaling

The UI no longer forced an immediate jump to RUNNING during these tests. Standby and live mV values were readable while signals were present (addresses AP-VVR-002 F-005 / AP-007 / AP-008 enhancement).

Severity: Info

Status: Improved — closed for this firmware on Rev 1 bench conditions

F-006 — Hardware Revision 2 not validated

Only Hardware Revision 1 was available. Batch-2 production uses revised trigger hardware. This report cannot claim Rev 2 trigger validation.

Severity: Medium (release scope)

Status: OPEN — blocking for Rev 2 engineering sign-off

4. Questions for Noxon

Q-001

In Relay Contact mode, why does a connected but inactive / floating bench supply (readout ~1.91–1.96 V) cause continuous triggering? What is the intended open-circuit and maximum leakage voltage for “no trigger”?

Q-002

Is attaching a voltage supply to the trigger input while the unit is in Relay Contact mode a supported configuration, an unsupported misuse, or a condition firmware should reject / ignore?

Q-003

Please confirm intended thresholds (mV or V) for Relay Contact vs 12/24 V modes, and whether 9 V must not fire 12/24 V mode on firm05-08 (observed). *(Partial answer received — see Iker-Trigger-Thresholds-2026-08.md: relay ≈1850 mV / clean ≈1950 mV; 12/24 INFO ≈2400 & 800 mV; applied ~6–8 V polarity-dependent. Open: why 9 V @ ~2700 mV INFO did not fire 12/24 this session — polarity?)*

Q-003a

Given clean contact ≈1950 mV and trip ≈1850 mV, what raised relay threshold do you recommend for short studio cables without losing long-cable detection?

Q-004

Does the “driver management” change in firm05-08 affect Hardware Revision 1, Revision 2, or both? (Asked separately to Iker; still needed for the revision log.)

Q-005

What minimum validation on Hardware Revision 2 does Noxon recommend before Batch-2 shipment, given this session was Rev 1 only?

5. Recommendations

R-001

Treat F-001 as an open critical issue. Do not declare Relay Contact validation complete on firm05-08 until Noxon explains and, if needed, corrects the inactive-supply false-trigger behavior.

R-002

Do not grant unconditional engineering release for Hardware Revision 2 Batch-2 units based solely on this Rev 1 session. Plan a short Rev 2 smoke of T-01–T-03, T-05–T-07, and the F-001 wiring case when a Rev 2 unit is available (or require Noxon's factory evidence).

R-003

Document operator guidance: until F-001 is resolved, avoid leaving a voltage supply connected to the trigger input while AutoPilot is in Relay Contact mode.

R-004

Send this report (and voltage table) to Iker with emphasis on Q-001 / Q-002.

6. Remaining Validation

- Reproduce F-001 with defined open / short / pull-down / disconnected reference setups
- Hardware Revision 2 repeat of critical matrix (T-01–T-03, T-05–T-07, F-001 case)
- T-04 studio noise / nearby motion (not available this session)
- T-10 startup current measurement (instrumentation)
- Iker written note on driver-management change (Rev 1 vs Rev 2)
- Alfredo / Noxon Rev 1 field-support policy (firmware vs board replacement)

Engineering Assessment

Dimension	Conclusion
Intentional trigger function (Rev 1, firm05-08)	PASS under tested fire counts
Idle spontaneous false-trigger watch (Rev 1)	PASS this session
Relay Contact electrical robustness	FAIL / OPEN (F-001 continuous false trigger)
Diagnostics / monitoring	IMPROVED (F-005)
Hardware Revision 2 Batch-2 release	NOT CLEARED by this report
Overall Validation Status	OPEN
Overall Functional Result	CONDITIONAL PASS

Shipment posture: Manufacturing may continue preparing the batch, but engineering approval to ship should remain conditional on (1) Noxon response / disposition of F-001, and (2) Rev 2 evidence or an explicit accepted risk decision by Performing Rigs that Rev 1 bench results plus Noxon factory test are sufficient.

Revision History

Rev	Date	Description
A	2026-08-06	Initial issue — firm05-08 / V01 R39 + R42 on Hardware Rev 1